

Hasan Al Saeedi

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science

Champaign, IL

Aug 2023 – May 2026

- Cumulative GPA: 3.60/4.0
- **Relevant Coursework:** Data Structures; Algorithms & Models of Computation; Systems Programming; Applied Parallel Programming; Distributed Systems; Communication Networks; Numerical Analysis; Cyber Security; Artificial Intelligence

SKILLS

Languages & Embedded: C/C++, Rust (no-std, embedded), Python, FreeRTOS, UART, GDB, OpenOCD (SWD), TCP/UDP, TLS, MQTT

Tools: Git/GitHub, GoogleTest, CMake, Linux, ROS 2, Docker, KiCad, Gazebo, Wireshark

PROJECTS

Embedded Rust on STM32F303 | In Progress

- Flashed and debugged on target over SWD with OpenOCD and GDB, using breakpoints, register/memory inspection, and single-stepping to analyze target's register usage.
- Building a register-level UART driver in Rust (no_std, PAC) for an STM32F303, configuring USART1 clocks, pins, and baud rate for 115200 8N1.
- Integrating with a Raspberry Pi 5 over serial to exchange data between a Linux host and bare-metal firmware.

ESP32 Plant Monitoring Node | In Progress

- Built a battery-powered FreeRTOS application on an ESP32 (ESP-IDF) that monitors soil moisture and air temperature/humidity, with separate tasks for sensor sampling, threshold evaluation, and Wi-Fi alerting.
- Sends Wi-Fi alerts via MQTT when moisture/temperature thresholds are exceeded, with a status LED for local indication.

Earthrover Mini+ LeRobot Integration

- Wrote a TCP-to-UART bridge daemon in C for an embedded Linux SoC (epoll, non-blocking I/O, termios raw 8N1) with client reconnect, connecting a Python host API to an STM32 motor controller.
- Implemented a Python host API for a 15-message binary UART protocol (CRC-16 framing, stream parser with resync, ack/retry, IMU/RPM telemetry decoding).
- Debugged dropped motor commands by isolating the fault with keep-alive ping/pong (link and CRC verified good) and tracing it to malformed packet fields.

F1 World Driver Championship Tracker | github.com/Hassoonu/F1WDCTrackerFFI

- Developed a Python script that uses custom C code as a Foreign Function Interface (FFI).
- Used OpenSSL and the Windows socket library to achieve secure network communication with an external API.

WORK EXPERIENCE

SIGRobotics

Project Lead

Champaign, IL

Aug 2025 – Jun 2026

- Led a 20-person team building an autonomous 1/10-scale racing stack on an NVIDIA Jetson Thor, using ROS 2 with perception, planning, and control nodes in C++ and Python.
- Won 1st place at the SeedStudio/NVIDIA/Hugging Face robotics hackathon, fine-tuning an NVIDIA GR00T imitation learning policy on bi-manual SO101 arms running on a Jetson Thor.
- Developed in a 5-person team using a Git/GitHub workflow with pull requests and peer code review.

University of Arizona – CAT Vehicle

Undergraduate Researcher

Tucson, AZ

Jun 2023 – Aug 2023

- Researched a method to verify physical message veracity in a connected autonomous vehicle system with attackers that have trusted credentials in a high-speed, safety-critical environment.
- Created PLEXE platooning simulations in Python and SUMO and managed changes with Git.